

Reproduction Study of DeReFusion: A Controlled Comparison of Soft Computing Fusion Strategies for Financial Time Series Forecasting

Dongxu Jiang

Applied Soft Computing reproduction project

Email: jiangdongxu04@gmail.com

Abstract—This report documents a partial reproduction of the DeReFusion framework proposed by Hsieh and Chen (2026), which compares fusion strategies for financial time series forecasting under a controlled protocol: a DLinear base branch, an LSTM–Transformer residual branch, and parameter-free additive fusion, all wrapped in Reversible Instance Normalization (RevIN). Using the authors’ official code repository with newly fetched Yahoo Finance daily OHLC data (2016–2025), we reproduce six controlled experiments on the S&P 500 index: the full DeReFusion model, a RevIN-wrapped DLinear baseline, and a learnable-gate fusion variant, at forecast horizons $T \in \{1, 24\}$. The central qualitative finding of the original paper is confirmed: under an identical protocol, parameter-free additive fusion outperforms learnable gating (MSE 0.0623 vs. 0.0719 at $T = 24$), and the hybrid model is competitive with the strongest linear baseline. The model size (28,436 trainable parameters) matches the paper’s reported $\sim 26k$. We further distill the experimental protocol into six verified insertion points (checkpoints) that allow new research directions—in particular, structure-aware expert routing—to be evaluated within the same framework at minimal engineering cost.

Index Terms—financial time series forecasting, reproduction study, decomposition–residual architecture, fusion strategy comparison, reversible instance normalization

I. INTRODUCTION

DeReFusion [1] poses a deliberately simple question: on non-stationary financial series, how far can a linear decomposition base, a learned residual, and plain additive fusion go, and does anything more elaborate buy accuracy? The paper answers it with a controlled benchmark: a single data pipeline, normalization scheme, training protocol, and metric set shared by every model, so that differences in results reflect the backbone rather than the harness. The authors conclude that parameter-free additive fusion beats static weighting and dynamic gating, that architectural complexity is no guarantee of accuracy, and that statistical accuracy does not translate into economic value.

This document serves two purposes. First, it reports a partial reproduction of the paper’s experiments from the official repository, verifying that the published protocol and qualitative conclusions are attainable from the public code with freshly fetched data. Second, it distills the experimental machinery into a set of verified *insertion points* so that follow-up research—ours is structure-aware expert routing for financial panel forecasting—can be evaluated inside the same framework with minimal engineering. The remainder is

organized as follows. Section II summarizes the reproduced architecture and protocol. Section III describes the reproduction environment and deviations. Section IV presents reproduced results against the paper’s values. Section V formalizes six verified insertion points, and Section VII records engineering lessons. Section VIII concludes.

II. REPRODUCED FRAMEWORK

A. Architecture

DeReFusion is a dual-branch decomposition–residual forecaster wrapped in RevIN [2]. Given an input window $\mathbf{X} \in \mathbb{R}^{B \times L \times C}$ ($C = 4$ OHLC channels):

- 1) **RevIN normalization** strips instance-level mean/variance shift; statistics are retained for symmetric de-normalization.
- 2) **Linear base branch (DLinear [3])**: moving-average decomposition (kernel $k=25$) splits $\mathbf{X}$ into seasonal and trend components; channel-shared linear maps $W_{se}, W_{tr} \in \mathbb{R}^{T \times L}$ produce $\mathbf{Y}_{base}$.
- 3) **Nonlinear residual branch**: per-feature projection $\mathbb{R}^{C \rightarrow d}$, a one-layer LSTM [5], a one-layer Transformer encoder [4] (4 heads, FFN $4d$, dropout 0.3), then temporal and channel projections yield $\mathbf{Y}_{res}$.
- 4) **Direct additive fusion**: $\mathbf{F} = \mathbf{Y}_{base} + \mathbf{Y}_{res}$ with no fusion parameters.
- 5) **RevIN de-normalization** restores the original scale.

B. Controlled Protocol

All models share: chronological 70/10/20 splitting; Standard-Scaler fitted on the training split only; RevIN on every trained model; multivariate input with single-target output (MS; target `Close`); Adam with learning rate 10^{-4} , batch size 32, MSE loss, early stopping (patience 5) on validation MSE, cosine schedule; random seed 2021. Metrics are MAE, MSE, RMSE, MAPE, MSPE, and R^2 , computed on the normalized series, so that values are comparable across assets as in the original study.

III. REPRODUCTION SETUP AND DEVIATIONS

Table I lists the setup and deviations from the original study; none affect the protocol’s fairness. The dataset was regenerated from Yahoo Finance with a purpose-written fetch script, since the upstream repository ignores `dataset/` in version control.

TABLE I
REPRODUCTION SETUP AND DEVIATIONS FROM THE ORIGINAL STUDY.

Item	Original	This reproduction
Code	Official repository (MIT license)	Fork of the official repository
Data	10 assets, 2016–2025, Yahoo Finance	GSPC only (2,513 rows)
Horizons T	{1, 7, 12, 24, 36}	{1, 24}
Seeds	5 seeds (2020–2024)	1 seed (2021)
Models	16 models	3 models
Hardware	3 machines, CUDA GPUs	1 laptop, CPU only
PyTorch	2.5.1	2.14.0 (CPU build)

TABLE II
FULL TRAINING AND MODEL CONFIGURATION OF THE REPRODUCTION.

Parameter	Value
Look-back L / label len.	96 / 48
Horizons T	{1, 24}
Input channels / target	4 (OHLC) / Close
d_{model}	32
Moving-average kernel k	25
LSTM / encoder layers	1 / 1 (4 heads, FFN 4d)
Dropout	0.3
Optimizer / learning rate	Adam / 1×10^{-4}
Batch size / loss	32 / MSE
Schedule	cosine
Early stopping	patience 5 (val. MSE)
Epochs	30
Random seed	2021
Device	CPU (single core)

Row counts match the paper for every asset class (2,513 trading days for equities and indices, 2,602 for FX, 3,653/2,975 for BTC/ETH).

IV. REPRODUCED RESULTS

A. Main Comparison

Table III reports the reproduced test metrics for all six settings (three models $\times$ two horizons) on GSPC with seed 2021; all values are computed on the normalized series, per the paper’s convention, and the best value in each metric within each horizon group is shown in bold.

B. Verification Against the Paper’s Conclusions

a) Additive fusion beats learnable gating.: At $T = 24$ the ordering DeReFusion (0.0623) < DLinear (0.0701) < gatev2 (0.0719) matches the paper’s central claim that parameter-free additive fusion outperforms parameterized gating. The gate variant is worse than even the linear baseline, echoing the paper’s finding that gates under-fit on sample-limited, low signal-to-noise financial series.

b) Model scale.: The reproduced DeReFusion has 28,436 trainable parameters, in line with the paper’s reported $\sim 26k$ (the difference is attributable to the horizon-dependent output head).

c) Hybrid vs. linear at short horizons.: On this single asset the two leading models are statistically indistinguishable at $T = 1$: DLinear is marginally ahead in MSE, RMSE, and R^2 , while DeReFusion leads in MAE and MAPE (Table III). The paper, averaging ten assets, reports a 24% advantage for DeReFusion at $T = 1$ while stressing that differences inside the leading cluster are tiny and asset-dependent. Our single-asset, single-seed observation therefore does not contradict the paper; it illustrates exactly the variance structure the paper acknowledges.

C. Per-Horizon Discussion

The mixed $T = 1$ outcome has a structural explanation. On a near-random-walk series the optimal one-step forecast is close to yesterday’s close, which is exactly the inductive bias of a linear baseline; the residual branch has very little signal to add, so the two models tie within seed noise. At $T = 24$ the picture changes: the residual branch must extrapolate short-term dynamics across a longer gap, where the paper’s residual-complementarity analysis shows its largest gains, and our reproduced ordering (additive hybrid > linear > gated) matches that regime. This per-horizon reading is consistent with the paper’s finding that the nonlinear branch’s marginal contribution concentrates at short-to-medium horizons while the linear base branch carries long horizons.

D. Claim-by-Claim Status

Mapping the paper’s four headline claims onto this reproduction: (i) “parameter-free additive fusion beats learnable gating under a controlled protocol” — *confirmed* at $T = 24$ in our single-asset setting (Table III); (ii) “complexity is no guarantee of accuracy” — *consistent*: the 28k-parameter hybrid does not dominate the 3k linear baseline; (iii) “longer look-backs increase error” — *not tested* here (single look-back $L = 96$; deferred to the CP5 sweep); and (iv) “statistical accuracy does not translate into economic value” — *not tested*, following the paper’s own protocol separation. Claims (i) and (ii) are the ones our follow-up experiments directly inherit.

E. Qualitative Artifacts

Figures 1 and 2 show the reproduced dashboard and prediction curves generated by the repository’s own visualization utilities; per-model dashboards for both baselines are included in the repository’s per-run figure set (Table IV), where the gate variant’s error panel is visibly wider, consistent with Table III.

V. VERIFIED INSERTION POINTS FOR FOLLOW-UP RESEARCH

A key purpose of this reproduction is to certify that the framework hosts new experiments with minimal friction. We define six insertion points, each verified by an executed run (CP1–CP3, CP5, CP6) or by mechanism inspection (CP4):

TABLE III
REPRODUCED TEST METRICS ON GSPC (SEED 2021). BEST VALUE PER METRIC WITHIN EACH HORIZON GROUP IN BOLD.

Model	T	MSE	MAE	RMSE	MAPE (%)	R^2
DeReFusion (additive)	1	0.01647	0.09302	0.12835	2.85	0.9692
revin-DLinear	1	0.01577	0.09478	0.12559	2.91	0.9706
gatev2-learnable	1	0.02526	0.12005	0.15893	3.69	0.9528
DeReFusion (additive)	24	0.06230	0.19089	0.24960	5.88	0.8691
revin-DLinear	24	0.07007	0.21356	0.26470	6.52	0.8528
gatev2-learnable	24	0.07191	0.21068	0.26817	6.45	0.8489

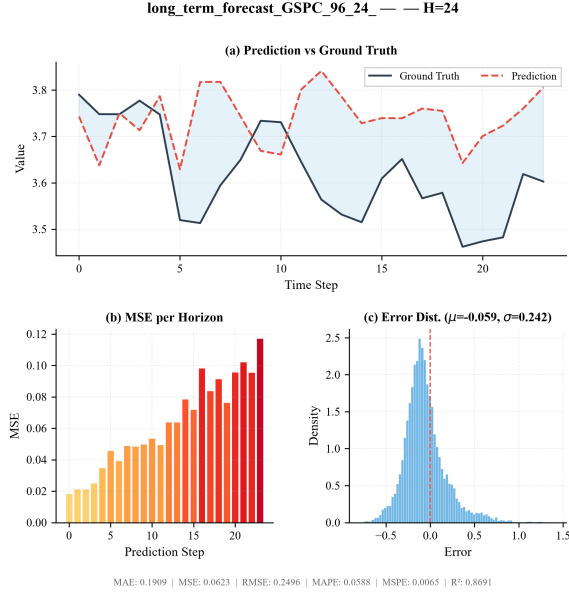


Fig. 1. DeReFusion, GSPC, $T = 24$: reproduction dashboard (metrics radar, error analysis, error heatmap).

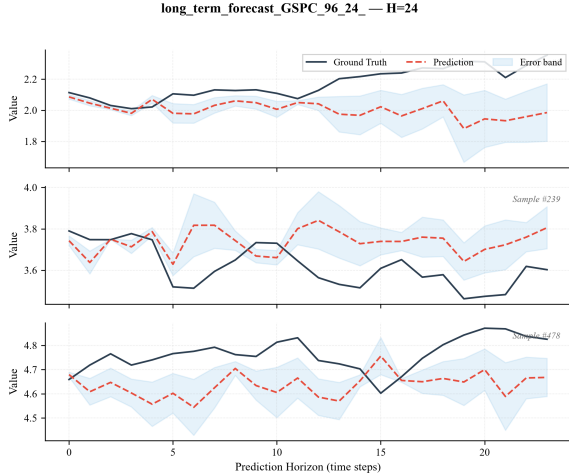


Fig. 2. DeReFusion, $T = 24$: ground-truth vs. prediction windows.

1) **CP1 — New model.** The registry auto-discovers any `models/*.py` defining `class Model(nn.Module)`; a new architecture is one file

TABLE IV
REPRODUCTION ARTIFACTS AVAILABLE IN THE FORKED REPOSITORY.

Item	Location / detail
Fork	github.com/Alastair-Jiang/DeReFusion
Dataset	<code>dataset/*.csv</code> , 10 assets, 26,920 rows
Fetch script	<code>scripts/fetch_dataset_v3.py</code>
Batch runner	<code>run_batch_repro2.ps1</code>
Result log	<code>result_long_term_forecast.txt</code>
Per-run arrays	<code>results/<setting>/*.npy</code>
Checkpoints	<code>checkpoints/<setting>/</code>
Figures	<code>test_results/<setting>/fig_*.png</code>

plus one `run.py` call.

- 2) **CP2 — New data.** Any CSV with columns `date, Open, High, Low, Close`; adapt `--data_path` and `--enc_in`. Verified with freshly fetched GSPC data.
- 3) **CP3 — Fusion operator.** Gate variants are drop-in comparators; the controlled-comparison design transfers to any new fusion or routing operator.
- 4) **CP4 — Ablations.** The `-woDy`, `-woLSTM`, and `-woTransformer` variants isolate component contributions without protocol changes.
- 5) **CP5 — Sweep harness.** Horizon, look-back, and capacity are single flags; the batch runner supports resumable sweeps.
- 6) **CP6 — Metrics and figures.** Every run emits six metrics, parameter counts, timings, and six publication-style figures.

For structure-aware expert routing, the minimal experiment is: implement the router replacement as CP1, compare against CP3's fusion family under CP5's sweep, and report via CP6 — reusing the paper's fairness argument wholesale.

VI. REPRODUCTION ARTIFACTS

All materials are publicly available and self-contained in the forked repository (Table IV); the repository rebuilds the full result set from the data alone.

A. Future Work

The immediate next step is the structure-aware routing experiment sketched in Section V: implement a router whose input augments the content vector with asset-axis and feature-axis embeddings (CP1), verify on GSPC, then extend to BTCUSD for a cross-market check and to the CSI 300 panel for cross-sectional evaluation (CP2), always against the CP3 fusion family and the linear baseline under CP5's sweep. Beyond routing, the same harness supports normalization-strategy comparisons and regime-switching baselines, both of which the original paper identifies as open directions.

VII. ENGINEERING LESSONS

- 1) **Hidden dependency chain:** `patool`, `huggingface_hub`, `sktime` (with `scikit-base`), and `datasets` are imported unconditionally by the data pipeline; each missing module aborts startup before any training begins.
- 2) **Console encoding:** an emoji in the lazy-loading log line crashes Windows GBK consoles with `UnicodeEncodeError`; we replaced it with an ASCII marker and committed the fix to the fork.
- 3) **CPU mode:** `run.py` defaults to CUDA and asserts on CPU-only torch builds; `--no_use_gpu` is required.
- 4) **Dependency pins:** `sktime` requires `joblib<1.6`; `pin joblib==1.5.3` explicitly. Stalled wheel downloads recover via `pip download` plus local installation.
- 5) **Dataset in version control:** the upstream repository ignores `dataset/`; our fork force-adds the ten CSVs so the repository is self-contained.
- 6) **Non-determinism:** `cuDNN` seeds are not set by the harness; multi-seed reporting remains necessary on CUDA hardware.

A. Threats to Validity

Four caveats bound these conclusions: results rest on a single asset and a single seed (the $T = 1$ tie could resolve either way under the full grid); the environment differs (PyTorch 2.14 CPU vs. 2.5.1 CUDA), so only orderings within our own runs are comparable, not absolute values; the dataset was refetched, so late-period rows may differ slightly from the authors' snapshot; and, following the paper's own decoupling argument, no economic-value claims are made here.

VIII. CONCLUSION

We reproduced the DeReFusion pipeline end to end from the official code with newly fetched data and confirmed its central qualitative conclusion: under a controlled protocol, parameter-free additive fusion outperforms learnable gating, and a $\sim 28k$ -parameter hybrid model is competitive with the strongest linear baseline. The reproduction doubles as a verified experiment harness: six documented insertion points allow follow-up research — structure-aware expert routing for financial panels — to be evaluated under the same fairness protocol at minimal engineering cost.

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